

[Juan Acosta](#)

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DATA SCIENCE & MACHINE LEARNING CANDIDATE | NLP & AI PROJECTS

Data science and machine learning candidate with hands-on experience building end-to-end ML and NLP systems, including LLM-powered applications and audio classification models. Strong foundation in Python, SQL, and deep learning, with experience designing data pipelines, training models, and implementing real-world AI workflows. Background in music and teaching brings strong communication, adaptability, and problem-solving skills. Interested in applied ML, AI-driven products, and data-focused roles.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Machine Learning & NLP: scikit-learn, TensorFlow, NLP, LLM workflows, model evaluation

Data & Analysis: pandas, numpy, data processing, feature engineering

Tools & Frameworks: Flask, LangChain, vector databases, AWS, Git

Web: HTML, CSS

Concepts: Machine Learning, Deep Learning, Statistical Analysis, Data Structures & Algorithms

TECHNICAL PROJECTS

[RAG Real Estate Chatbot \(LLM Application\)](#)

Built an end-to-end LLM-powered system to automate property inquiries and route users to agents via WhatsApp.

- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain
- Integrated WhatsApp messaging via Twilio API for real-time user interaction
- Built a Flask backend to orchestrate LLM responses and manage application logic
- Developed and fine-tuned a BERT-based intent classifier to detect "contact agent" requests
- Implemented semantic search using a vector database for efficient information retrieval
- Designed multi-step context handling to support conversational workflows and improve user experience

[Distance Matrix System \(Google Maps API\)](#)

Built a data pipeline to generate and analyze travel distance matrices using external APIs.

- Designed API integrations to collect travel time and distance data at scale
- Processed and structured data using Python and pandas for downstream analysis
- Developed reusable functions for automated data retrieval, transformation, and storage
- Enabled SQL-compatible outputs to support analytical workflows

[Urban Sound Classification \(Machine Learning\)](#)

Built a supervised learning system to classify environmental sounds using engineered audio features.

- Processed audio data from UrbanSound8K using Librosa (MFCCs, spectral features)
- Engineered features and trained multiple models including Logistic Regression, XGBoost, and CNNs
- Evaluated model performance and achieved ~70% accuracy with XGBoost as the top-performing model
- Compared model architectures to optimize predictive performance

[Speech Recognition Model \(Deep Learning\)](#)

Developed a deep learning system to identify speakers from audio samples.

- Extracted and engineered Mel-spectrogram and MFCC features from raw audio data
- Conducted exploratory data analysis to identify key predictive features
- Designed and trained neural network architectures including Bidirectional LSTM models
- Achieved 85% test accuracy with optimized model configuration

EDUCATION

NYU Tandon School of Engineering, New York, NY

01/2025 - 06/2025

Bridge to Computer Science Program	
Flatiron School, New York, NY Immersive Data Science Bootcamp program	11/2023 - 03/2024
SUNY Purchase, Purchase, NY Master of Music in Jazz Performance	08/2015 - 05/2017
Webster University, St. Louis, MO Bachelor of Music in Jazz Studies	08/2010 - 05/2014

EXPERIENCE

Café Attendant, Millennium Hilton — New York, NY	10/2024 – Present
- Deliver high-volume customer service in a fast-paced, operational environment - Manage transactions, order flow, and customer interactions under time constraints - Maintain consistency and efficiency in service delivery during peak demand	
Barista, Starbucks , New York City, NY	08/2022 - 10/2024
- Provided customer service in a fast-paced, high-volume retail environment - Managed beverage preparation workflows with consistency and attention to detail - Maintained operational standards and cleanliness during peak hours	
Music Teacher, Colegio Menor , Quito, Ecuador	01/2019 - 02/2020
- Restructured music theory and audio technology curriculum - Wrote material for teaching scales, chords, and rhythm - Taught music technology skills like recording, mixing, sound editing, and sampling	
Freelance musician and instructor, Self-employed , New York City, NY	08/2015 - 05/2018
- Performed in multiple styles in New York City and St. Louis - Taught private music lessons to children and adults	